

Lee Kai Yi Zachary

• lkyzachary@gmail.com • +65 8142 2748 • DOB: 15 Nov 2002

Portfolio: <https://zachyboiii.github.io/zac-portfolio/> | LinkedIn: <https://www.linkedin.com/in/zachary-lee-ky/>

Self-motivated and results-driven aspiring software engineer with interests in full-stack development and AI. Proven leadership through project work, student initiatives, and volunteer coordination. Available from May 2026 to Aug 2026.

WORK EXPERIENCES

Stajenn Technologies Pte Ltd Jan 26 – May 26
Incoming AI/ Full Stack Development Intern

Singapore University of Technology and Design (SUTD) May 25 – Aug 25
Teaching Assistant 10.020 Data Driven World

- Assisted with two classes supporting learning in Python programming, data manipulation using Pandas/NumPy, and foundational algorithms through consultations and debugging help.
- Graded assignments and project components, providing technical feedback to improve code quality and logical problem-solving.
- Collaborated with faculty to refine course materials and enhance student learning outcomes.

EDUCATION

Singapore University of Technology and Design (SUTD) Sep 23 – Present

- Bachelor of Engineering (Computer Science and Design), expected date of graduation: May 2027
- GPA 4.65/5.0 (Honors with Highest Distinction), SUTD Global Distinguished Scholarship Holder
- Relevant coursework: Computational Thinking for Design, Data Driven World, Data Structures and Algorithms, Computation Structures, Information Systems and Android Programming, Database Systems and Big Data, Computer Systems Engineering, Elements of Software Construction

University of Waterloo Sep 25 – Dec 25

- Relevant coursework: Introduction to Machine Learning, Introduction to Artificial Intelligence, Human Computer Interactions

PROJECTS

Chess AI Engine | Reinforcement Learning Agent Jan 26 – Jan 26

- Developed a chess AI engine using **PyTorch** and **deep reinforcement learning**, training a **neural network** through self-play to evaluate board positions and predict move probabilities
- Implemented **reinforcement learning** training pipeline with policy and value networks, optimizing move selection through iterative self-play and experience-based learning

Storehouse | Volunteer Organization Delivery Management System Dec 25 – Jan 26

- Developed key features for a full-stack delivery management platform serving a volunteer organization, enabling efficient coordination of deliveries to beneficiaries
- Built **RESTful** backend services with **Node.js** and **Express.js**, implementing full **CRUD operations** in **PostgreSQL** for delivery management and scheduling
- Created responsive frontend features using vanilla **JavaScript**, **HTML**, and **CSS**, including an interactive Google Calendar-style delivery view, delivery creation forms, and edit/delete functionality

Singapore University of Technology and Design | 50.003 Elements of Software Construction May 25 – Aug 25

- Built a full-stack hotel booking platform with Ascenda Loyalty featuring advanced search with **lazy loading**, real-time sorting/filtering, and interactive map visualization using **React.js**
- Architected **RESTful** backend services with **Express.js** and **MongoDB**, implementing secure user authentication, session management, and optimized API endpoints for hotel data retrieval
- Achieved **90%+ test coverage** through comprehensive **Jest unit tests** and **Playwright end-to-end testing**, ensuring application reliability across critical user flows

TruthieGO | DSTA BrainHack 2025 – Champion, CODE_EXP Competitive Category May 25 – Jun 25

- Engineered the core game logic for TruthieGO, a 2D educational board game, leveraging the **Godot engine** and GDScript to establish a modular, maintainable architecture supporting multiple game modes and interactive mini-games.
- Integrated secure and scalable backend services by connecting **Firebase (NoSQL)** for robust user authentication and session management, enabling real-time progress tracking across devices.

Singapore University of Technology and Design | 50.001 Information Systems & Android Programming Jan 25 – Apr 25

- Engineered Vroom, an **Android** application connecting learner drivers with private instructors, featuring real-time booking, scheduling, in-app chat, and interactive quizzes.
- Designed scalable architecture using Java OOP principles (MVC pattern, inheritance, interfaces) and Firebase Realtime Database for live booking updates and notifications

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, SQL, HTML/CSS, Tailwind CSS
- **Frameworks & Libraries:** React.js, Node.js, Express.js, Spring Boot, Streamlit, Pandas, NumPy, Matplotlib, Scikit-Learn, PyTorch
- **Testing:** Jest, Playwright
- **Tools & Platforms:** Git, Docker, MongoDB, Postgre SQL, Firebase, VS Code, Android Studio, Godot Engine, Lucid HDL, Hadoop, Apache Spark, Rhino, Figma
- **Methodologies:** REST API Design, Agile, Object-Oriented Programming

LEADERSHIP AND VOLUNTEERING

New Creation Church

Aug 22 – Present

Lead IC, Games – Kids Camp Volunteer

- Consistently led the planning and execution of engaging, age-appropriate game activities for large-scale kids camps with attendance exceeding 600 children per event.
- Managed and motivated volunteer teams across multiple camps, ensuring seamless coordination of game logistics, setup, and safety protocols.
- Acted as primary liaison with external vendors for equipment and operational support, guaranteeing timely and efficient resource delivery.

SUTD Freshman Orientation

May 24 – Sep 24

Orientation Group Leader Head

- Responsibilities included event planning and people management
- Leadership, teamwork and public speaking
- Mentored and integrated incoming SUTD Freshmen students

Volleyball Club

Sep 23 – Jan 25

Secretary

- Represented Singapore University of Technology and Design in the Singapore University Games both in 2023 and 2024
- Collaborated with other clubs such as SUTD Tchoukball Club to organize inter-club camps to expose members from both clubs to new experiences
- Responsibilities included managing club attendance, logistics and administrative matters
- Organized and managed club booth at school events such as Open House, Night Fiesta and Fifth row showcase